IJARSCT
ISSN (Online) 2581-9429
International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)
International Open-Access, Double-Blind, Peer-Reviewed, Refereed, Multidisciplinary Online Journal
Impact Factor: 7.67
Volume 5, Issue 5, March 2025

Energy Saving in Classroom using IoT

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Abstract: *This proposed system focuses on saving energy by using smart technology to control appliances like lights and fans automatically. The system uses motion sensors to detect if people are in a room. If the room is empty, the system turns off the lights and fans, conserving energy that would otherwise be inefficiently used.*

Although this system is designed for classrooms, it can be used in many other places like offices, homes, and public spaces. In offices, it can manage meeting rooms and workspaces, only turning on lights and fans when needed. At home, it can reduce electricity use by controlling lights in common areas. Public places like libraries and airports can also use this system to save energy when areas aren't busy. It's easy to set up in existing buildings and can be scaled up for larger spaces.

The technology uses PIR sensor to detect motion, once the motion is detected then camera starts to capture the video, the video is subdivided into frames by using according to the number of lights and fans in that room. After detecting motion in particular, the frame microcontroller gives a signal to relay and lights and fans in that area will be turned on.

By saving energy based on whether people are present, this system helps lower electricity bills and reduces carbon footprint which protects the environment..

Keywords: IoT, Camera-Based Detection, Smart Classroom, Automation, Energy Conservation

I. INTRODUCTION

Energy wastage is a significant concern in both residential and commercial areas, where inefficient management of lighting often leads to excessive energy consumption. Many buildings rely on traditional switches and timers, resulting in lights and fans remaining on in unoccupied rooms, which drives up energy bills and increases carbon footprints.

As the demand for sustainability grows, there is a pressing need for innovative solutions to minimize energy waste. Smart Energy Management System addresses this challenge by using a camera and PIR (Passive Infrared) sensor to detect a person's presence in a room. This system automatically turns on and off lights and fans based on real-time presence, enhancing comfort while significantly reducing energy consumption. By integrating smart home technology, this project promotes a more sustainable approach to energy use in modern living and working environments.

The proposed system aims to do the following actions:

- Determine the presence of a human in a room.
- Determine whether the devices in a room are turned ON or OFF.
- Control the devices by turning them ON/OFF based on human detection.
- Continuous monitoring of a room for entry/exit by any human.

II. RELATED WORK

IoT-based systems are a central theme in existing research, enabling real-time monitoring and control of appliances. Microcontrollers like the esp32 are frequently utilized due to their low power consumption and Wi-Fi capabilities These microcontrollers serve as the central hub, connecting to a Wi-Fi network and controlling relays that operate lights and fans. The esp32 cam module receives signals from sensors and sends commands to turn devices on or off based on occupancy. The integration of such microcontrollers facilitates communication and control within the systems.

Motion detection is another critical component, commonly achieved through PIR (Passive Infrared) sensors and cameras. PIR sensors detect motion by sensing infrared radiation emitted by people. Cameras, on the other hand, offer

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more advanced motion detection through image processing, which enhances the accuracy of occupancy detection. Integrating both PIR sensors and cameras improves the reliability of occupancy detection

Automation of lighting and appliance control is a key objective, aiming to minimize energy consumption by ensuring energy is used only when needed. By automating device control based on occupancy, significant energy savings can be achieved. The automated systems are designed to address the limitations of traditional systems that rely on manual controls or timers which often lead to energy waste.

These systems emphasize the importance of being practical and sustainable for modern smart classroom applications. They are also designed to address issues with traditional systems that rely on manual control or timers, which can lead to energy waste. The systems aim to solve this problem by using smart technology that can detect when a room is empty and automatically turn off lights and fans, thus helping to save energy.

The Smart Energy Management System specifically aims to address inefficient energy usage, particularly in classrooms, and provides solutions by automating the control of electrical devices based on real-time occupancy. The system combines PIR sensors and cameras to detect human presence. When occupancy is detected, the system activates relay modules to turn on lights and fans. When the classroom is vacant, the system automatically turns off these devices after a set delay. The system also allows for manual overrides.

The system's architecture includes:

A sensor layer with PIR sensors and cameras.

•A processing layer with the esp32 cam module microcontroller.

•A control layer with relay modules.

•A communication layer using Wi-Fi and HTTP protocols.

The design incorporates a user-friendly interface that allows facility managers and teachers to monitor and control the system, including manual override capabilities when needed.

III. IMPLEMENTATION

The Smart Energy Management System is implemented using a combination of hardware and software components to automate the control of lights and fans in classrooms. The hardware includes PIR sensors that detect motion via infrared radiation, cameras that capture visual data for enhanced occupancy detection, microcontrollers (esp32) that used to capture images and act as the central processing unit, relay modules that act as electronic switches to control the lights and fans, and Wi-Fi modules for wireless communication. Software components consist of programming languages (Python, JavaScript), and a database (such as MySQL) to store energy consumption logs. The system works by using the sensors to detect motion and occupancy, then processing the data to determine if the lights and fans need to be turned on or off. A user- friendly interface provides monitoring and override capabilities. The system is designed for real-time processing, automated control, scalability, user-friendliness, modularity, energy efficiency, and reliability. The implementation is tested using unit, integration, systems, and acceptance testing to verify the functionality of each module.

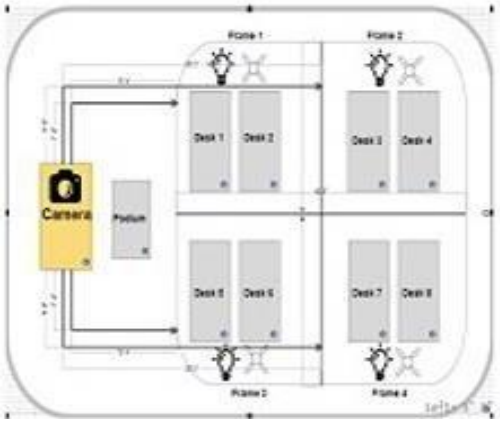


Fig. 1. A Representation of the Proposed Classroom Environment [1]

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Abbreviations and Acronyms

IoT: Internet of Things
PIR: Passive Infrared
ESP32: A low-cost Wi-Fi microcontroller
HTTP: Hypertext Transfer Protocol
TCP/IP: Transmission Control Protocol/Internet Protocol

IV. COMPONENTS

The Smart Energy Management System utilizes several key components to achieve automated control of lights and fans in a classroom. The system's hardware components include PIR sensors, which detect motion by sensing changes in infrared radiation, signaling the presence of people Camera of esp32 cam module is also included, which can optionally be used for enhanced motion detection and occupancy recognition using image processing techniques The microcontroller, an esp32, acts as the primary controller, receiving signals from sensors, processing data, and sending commands to turn devices on or off The microcontroller is also responsible for handling network communication. Additionally, relay modules are used as electronic switches that allow the microcontroller to control high- power devices such as lights and fans. A Wi-Fi module (esp32 cam module) facilitates wireless communication between the system and external devices for remote monitoring and data logging.



Fig.2. Four Channel Relay Module



Fig. 3. The ESP32 CAM Wi-Fi

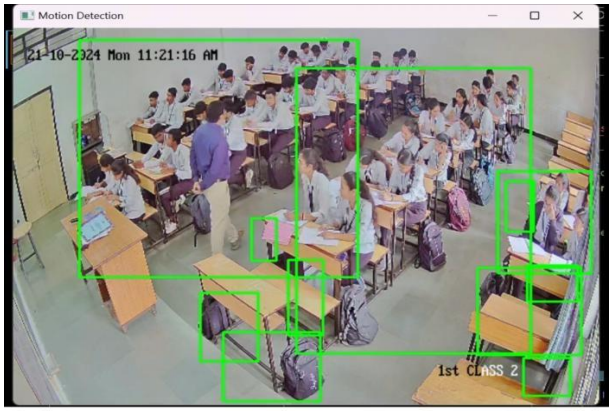


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The system's software components are equally important for its functionality. The system uses an operating system, such as Windows 11 for development. Programming languages, primarily Python with libraries like OpenCV for image processing, are used for device control, sensor data processing, and database interfacing. JavaScript is used for building the web interface. A database, such as MySQL, stores data related to energy consumption logs, user activity, and scheduling preferences. These components work in tandem to enable the system to detect occupancy, control devices, and provide remote monitoring capabilities.

V. RESULTS

The Smart Energy Management System is designed to automatically control lights and fans in classrooms based on occupancy, resulting in several key benefits after implementation. The system will provide automated energy management, reducing unnecessary energy consumption by turning off lights and fans when a room is vacant. This leads to reduced electricity bills and a lower carbon footprint, contributing to sustainability goals. The system also offers real-time monitoring and control capabilities through a user- friendly interface, allowing facility managers or teachers to view energy usage patterns, adjust settings, and manually control devices when needed. The implementation includes robust testing procedures including unit, integration, systems and acceptance testing to ensure reliability. Finally the system also provides data logging and analysis capabilities for optimizing energy usage and identifying areas for improvement.



VI. CONCLUSION

This paper presents a successful implementation of an energy-efficient classroom automation system. By utilizing the ESP32 CAM Wi-Fi module, the system intelligently controls appliances such as lights and fans based on real-time detection of human presence. The division of the classroom into distinct segments further enhances energy efficiency, as only the appliances in occupied areas are activated. This innovative approach demonstrates significant potential for reducing energy consumption and associated costs, making it a practical and sustainable solution for modern educational institutions. The proposed system stands as a promising advancement in energy conservation technology.

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